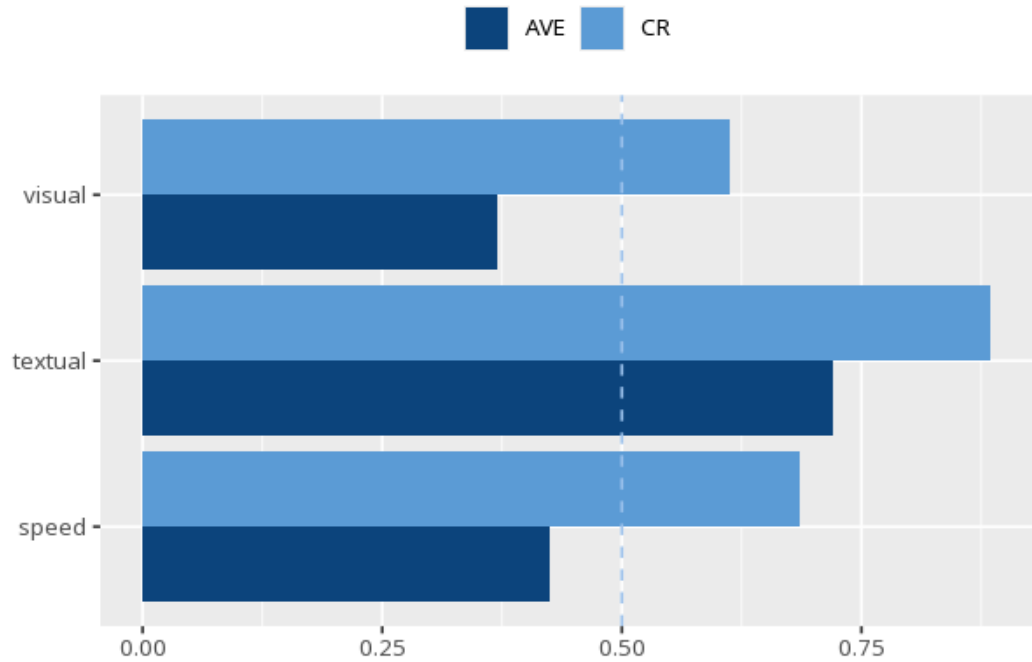


1 Average variance extracted (AVE)

Construct	AVE	CR	ω	α
visual	.37	.61	.61	.63
textual	.72	.89	.89	.88
speed	.42	.69	.69	.69

	visual	textual	speed
visual	.61		
textual	.46	.85	
speed	.47	.28	.65

	visual	textual	speed
visual			
textual	.38		
speed	.39	.28	



AVE (Average Variance Extracted) is the average share of item variance a construct captures; $\geq .50$ means items converge on their construct (Fornell & Larcker, 1981). CR (Composite Reliability) $\geq .70$ indicates acceptable construct reliability. ω (McDonald's omega) is the preferred headline reliability coefficient; α (Cronbach's) is shown as a secondary/legacy coefficient. For discriminant validity, Fornell–Larcker (Table 2): each construct's $\sqrt{\text{AVE}}$ (bold diagonal) should

exceed its correlations with other constructs. HTMT (Table 3): values $< .85$ support discriminant validity for conceptually distinct constructs — use HTMT as the primary criterion and Fornell–Larcker as a legacy check. AVE $< .50$ may still be acceptable when CR $> .60$ (F&L caveat), but treat this as a caution, not a lenient pass. All cutoffs are heuristics, not pass/fail gates. Applies to reflective constructs only.

Convergent validity was not fully supported, all AVE $\geq .50$ and CR $\geq .70$; discriminant validity held, all HTMT $< .85$.

Statistical basis: AVE (average variance extracted) convergent validity, $\geq .50$ (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." *Journal of Marketing Research*, 18(1), 39-50.); CR (composite reliability) $\geq .70$ acceptable (Nunnally, J. C. (1978). *Psychometric Theory* (2nd ed.). McGraw-Hill.); CR $\geq .70$ threshold, independent corroboration (Bagozzi, R. P., Yi, Y. (1988). "On the evaluation of structural equation models." *Journal of the Academy of Marketing Science*, 16(1), 74-94.); HTMT $< .85$ discriminant validity (primary criterion) (Henseler, J., Ringle, C. M., Sarstedt, M. (2015). "A new criterion for assessing discriminant validity in variance-based structural equation modeling." *Journal of the Academy of Marketing Science*, 43(1), 115-135.); Reliability preference (ω over α) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." *Psychological Methods*, 23(3), 412-433.).